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B.Tech. Degree IV Semester Examination April 2018

SE 15 - 1402 HEAT AND MASS TRANSFER OPERATIONS (2015 Scheme)

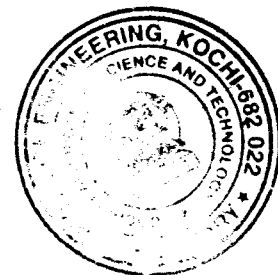
Time: 3 Hours

Maximum Marks: 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) How does thermal conductivity for solids, liquids and gases depend upon temperature?
- (b) How is heat transfer coefficient defined? What is the physical significance of 'film thicknesses'? Explain Newton's law of cooling.
- (c) How is absorptivity, reflectivity and transmissivity important in radiation?
- (d) What is fouling factor? Discuss its importance in design of heat exchangers.
- (e) For a solution, prove $D_{AB} \approx D_{BA}$.
- (f) Obtain the expression for Fick's law and also mention the conditions under which it is valid.
- (g) Obtain the expression for molar flux of liquids for steady state diffusion of A into non diffusing B.
- (h) State Raoult's law and Henry's law and why they are significant.
- (i) What is the physical significance of HTU and NTU?
- (j) Draw and explain working of a ballmann extractor.



PART B

(4 × 10 = 40)

- II. (a) An annealing chamber has a composite wall made of a 17 cm thick firebrick layer ($k = 1.1 \text{ W/m}^\circ\text{C}$) and a 13 cm thick ordinary brick layer ($k = 0.70 \text{ W/m}^\circ\text{C}$). The inside and outside surface temperatures of the wall are 400°C and 45°C respectively. Calculate the heat loss from 25 m^2 furnace wall. Also, determine the temperature between the ordinary brick and the fire brick layers. (5)
- (b) A 1 kW electric room heater has a coil of Nichrome wire of diameter 0.574 mm and electrical resistance of 4.167 ohm/m . If the temperature of the room remains constant at 21°C and the average heat transfer coefficient at the wire surface is $100 \text{ W/m}^2^\circ\text{C}$, calculate the time required for the heating coil after it is switched on to reach 63% of its steady state temperature rise. Assume that the wire offers negligible heat transfer resistance. The density of the material of the wire is 8920 kg/m^3 , and its specific heat is $384 \text{ J/kg}^\circ\text{C}$. (5)

OR

- III. (a) State and discuss: (6)
 - (i) Reynolds Analogy.
 - (ii) Prandtl analogy.
 - (iii) Colburn analogy.
- (b) Discuss the concept of boiling in heat transfer. Explain the various regimes involved. (4)

(P.T.O.)

- IV. What are the various factors involved in the design of heat exchangers? (10)
Draw the schematic of a single pass and multi pass heat exchanger.

OR

- V. (a) What is meant by radiation heat transfer? Discuss the concept of black body. (6)
Explain the laws of radiation.

- (b) Discuss the different types of evaporators and the method of feeding multiple effect evaporators. (4)

- VI. Discuss the different theories of mass transfer. (10)

OR

- VII. Design a packed tower with respect to: (10)

- (i) Individual mass transfer coefficient.
- (ii) Overall mass transfer coefficient.
- (iii) Height of transfer units.

- VIII. Explain and derive the McCabe Thiele method for design of a distillation column. (10)

OR

- IX. Discuss: (10)

- (i) Simple Leaching.
- (ii) Sprays dryers and its industrial application.
